

Yassine ABASSI

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EDUCATION

Polytechnique Montréal

Bachelor's in Computer Engineering

Montreal, QC

Aug 2022 – May 2026

EXPERIENCE

Software Quality Assurance Intern

May 2024 – Sep 2024

UPTOTEST

Ariana, Tunisia

- Designed and automated end-to-end functional test suites for the KOORS web app using Cypress with Cucumber/Gherkin in a BDD workflow.
- Translated product requirements into testable user stories and acceptance criteria, managed defects and traceability in Jira across sprints.
- Participated in Agile Scrum ceremonies to align QA priorities with development milestones and release timelines.

Private Tutor

Aug 2021 – Present

Self-employed

Montreal, QC

- Tutoring Mathematics, Physics, and Programming for college and university students.

PROJECTS

Hydro-Québec Weather Forecasting Platform | *Angular · .NET · EF Core · GDAL · Leaflet · SignalR · Docker*

- Built a full-stack weather visualization platform with Angular and ASP.NET Core for multi-model weather display with interactive geospatial mapping and real-time updates via SignalR.
- Implemented GDAL raster processing and Leaflet cartographic rendering; designed RESTful APIs alongside an SQLite database, containerized the application with Docker and set up a CI/CD pipeline for automated testing and deployment.

3D Interactive Portfolio | *React.js · Three.js · Upstash Redis*

[Live Demo](#) 

- Crafted a 3D portfolio with an interactive room featuring a functional OS, custom camera rigging, and state-driven transitions across mobile and desktop, expanded the experience with terminal emulation, procedural art, and embedded mini games.
- Integrated a Gemini-powered chatbot assistant via a server function with Upstash Redis-backed rate limiting.

SSH Terminal Portfolio | *Go · Bubble Tea · Lip Gloss · Wish · Docker · Fly.io*

[Live Demo](#) 

- Built an SSH-based interactive terminal portfolio using Go and the Charm ecosystem (Wish, Bubble Tea, Lip Gloss), featuring animated splash screens, keyboard navigation, and dynamic theme switching.
- Deployed on Fly.io with Docker containerization, persistent SSH host key management via mounted volumes, and automated CI/CD through GitHub Actions.

Corporate LinkedIn Translator | *Chrome Extension · Node.js · Express · Docker · Fly.io*

[Github](#) 

- Built and published a Chrome extension that uses Google Gemini AI to detect and translate LinkedIn corporate buzzwords into satire plain language with inline tooltip annotations.
- Architected a full backend API on Fly.io with rate limiting, graceful error handling, and dynamic DOM injection via a MutationObserver to handle LinkedIn's React SPA structure.

WildGuard – AI Wildlife Monitoring System | *Python · YOLOv8 · OpenCV · Flask · Socket.IO · Twilio*

[Github](#) 

- Built an AI-powered ecological surveillance platform for anti-poaching and endangered species protection, with real-time detection updates via Socket.IO and automated SMS alerting through Twilio.
- Implemented a YOLOv8 detection model capable of identifying 50+ species, integrated with OpenCV for video processing.

Multi-Robot Exploration System | *Python · ROS2 · Gazebo · Docker*

- Designed a coordinated multi-robot exploration system with AgileX Limo robots.
- Implemented navigation and inter-robot coordination in Python while containerizing software modules using Docker.

Embedded Obstacle Detection System | *C++ · ATmega164A*

- Programmed a fully autonomous mobile robot in C++ on an ATmega164A microcontroller, implementing low-level hardware control including PWM motor driving, sensor polling, and interrupt-driven decision logic.
- Designed obstacle detection and path-following algorithms using real-time sensor feedback.

TECHNICAL SKILLS

Languages: C, C++, C#, Java, Python, JavaScript, TypeScript, Go, Kotlin, Swift, VHDL, Assembly

Web & Mobile: Angular, React.js, Node.js, NestJS, Express.js, WebSockets, SignalR, .NET, Three.js

Data & APIs: PostgreSQL, MongoDB, SQLite, Firebase, Retrofit, OkHttp, EF Core, GDAL, RxJS, Upstash Redis

DevOps & Tools: Git, Github Actions CI/CD, GitLab, Docker, Jira, Fly.io, Qt, Cypress, Jasmine, Vivado, Wireshark

Robots & Boards: Zynq7000, ATmega164A, Pynq-Z1, Basys3 Artix-7, AgileX Limo

ADDITIONAL

Languages: English, French, Arabic (fluent) | German (intermediate)

Work Authorization: Authorized to work in the U.S. and Canada (dual citizen)